Marketing Analytics

Adaptive Recommendations and Marketing Budgets

By Nathan Yang

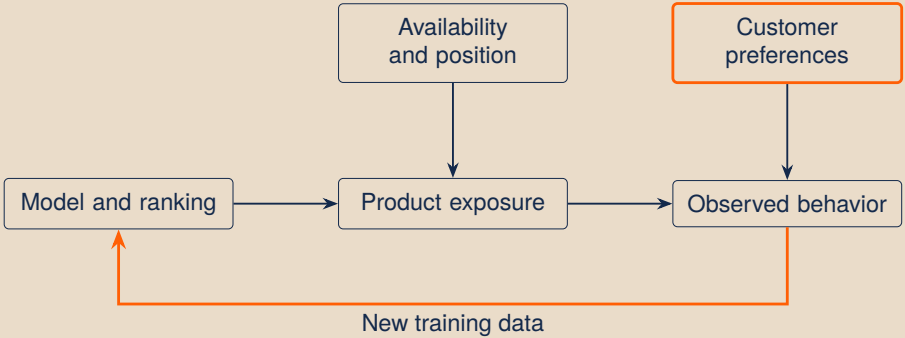
Session 13



An Adaptive Policy

- A model selects exposure using existing interaction histories.
- Customers respond to the displayed options.
- Retraining incorporates those selected responses.

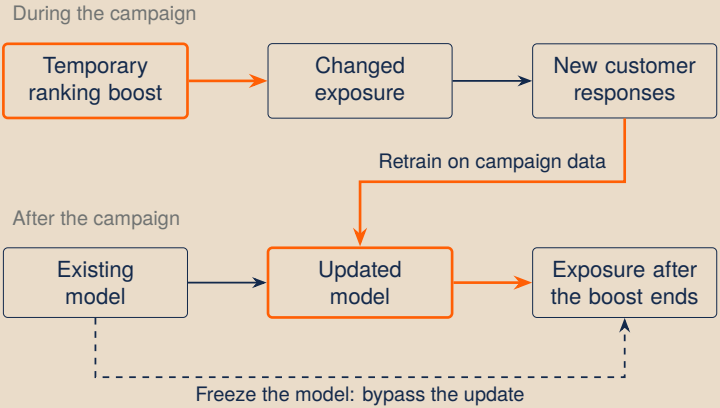
Selected Exposure Creates a Feedback Loop



Interaction data reflect both preference and opportunity.
A feedback loop can reinforce initially favored items.

The diagram describes a policy mechanism rather than an estimated effect.

The Evaluation Horizon Includes Learning



A temporary intervention can persist through subsequent model updates.

Customer Goals and Platform Objectives

- Clicks reveal a response within an offered environment.
- Considered preferences can differ from immediate engagement.
- Measure satisfaction, unwanted repetition, and customer-stated goals.

A List Objective

A list combines item relevance with category variety.

$$J(L) = \sum_{j \in L} s_j + \lambda K(L)$$

The variety weight λ is expressed in the same units as the relevance score.

Two Three-Item Lists

List	Relevance	Categories	Value at 0.30
A, B, C	2.50	2	3.10
A, C, D	2.30	3	3.20

Synthetic scores. The lists tie at a variety weight of 0.20.

Model-Based and Experimental Evidence

- A fitted choice model can compare alternative recommenders.
- A randomized perturbation can test predicted response patterns.
- Engagement, retention, revenue, and welfare are distinct outcomes.

Marketing-Mix Modeling

- Weekly channel spending can have delayed effects.
- Response can saturate at high exposure.
- Spending decisions can respond to anticipated demand.

Carryover

A geometric accumulated-exposure measure retains earlier spending.

$$a_t = s_t + \rho a_{t-1}, \qquad 0 \leq \rho < 1$$

A 100-unit pulse with $\rho = 0.5$ gives 100, 50, 25, and 12.5 in successive periods.

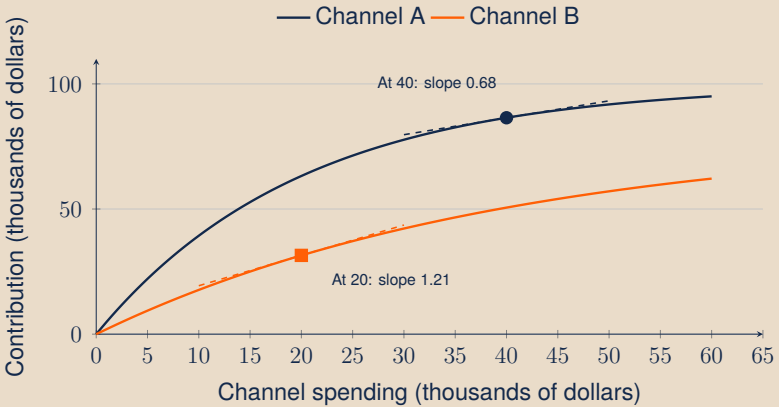
Saturation and the Next Dollar

A concave response has declining marginal contribution.

$$R(s) = A[1 - \exp(-s/b)], \qquad R'(s) = \frac{A}{b}\exp(-s/b)$$

Budget transfers depend on marginal response at the current allocation.

Average and Marginal Returns Can Disagree



Synthetic known response curves. Total spending is fixed at 60 thousand dollars.

The Fixed-Budget Optimum

Solve equal marginal returns for the imposed curves.

```
from math import exp, log
budget = 60
a = (log(2.5) + budget/40) / (1/20 + 1/40)
b = budget - a
ma = 5 * exp(-a/20)
mb = 2 * exp(-b/40)
assert abs(ma-mb) < 1e-12
```

The allocation is about 32.22 to A and 27.78 to B, in thousands of dollars.

Coding Exercise: A Policy Objective

- Enumerate three-item lists at several variety weights.
- Solve and independently grid-check the fixed-budget allocation.
- Report the assumptions that change the preferred action.

Attention and the Optimization Target

- A free service can require customer time and attention.
- More exposure can benefit advertisers while burdening customers.
- Evaluate voluntary use and customer outcomes alongside engagement.

Policy Discussion

- What changes when the recommender learns during the experiment?
- Which outcome measures the customer’s stated goal?
- How much uncertainty in response slopes would change the budget?